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**Zhiling Bei and Katrijn Gielens**

Abstract

China is the world's second largest retail market and is expected to surpass the United States in 2020. This makes China a must-win market for global and local firms. However, manufacturers often find it hard to reach Chinese consumers due to China's institutional and infrastructure weaknesses. The advent of online third-party marketplaces provides manufacturers with a new model to tackle these challenges. The question remains, however, whether and to what extent joining online third-party marketplaces in China can generate market value. Using an event study methodology, the authors assess the net impact of online marketplace adoption in China on manufacturers' expected profitability. Although the abnormal returns are, on average, significant and positive, stock markets recognize that not all manufacturers can reap benefits. To delve deeper into this variation, the authors relate the abnormal returns to various sources of manufacturers' market knowledge and marketing capabilities. They find that less diversified manufacturers with marketplace experience are expected to achieve greater gains, irrespective of their origin. However, the effects of marketing strengths diverge for foreign and local firms.

Keywords

the Chinese market, digital marketplaces, financial event study, online retailing

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China is the world's second largest retail market and is expected to surpass the United States in 2020. This makes China a must-win market for global and local brand manufacturers (Edge 2019). Manufacturers, however, often find it hard to reach, interact, and transact with consumers in China due to the institutional voids and infrastructure weaknesses they face. Whereas institutional voids refer to the failure of market-supporting and contract enforcement institutions to facilitate efficient exchange between firms (Khanna and Palepu 1997), infrastructure weaknesses pertain to the lack of robust and dependable distribution and logistics networks (Jung, Ugboma, and Liow 2015). To overcome such voids and weaknesses, manufacturers can typically exploit various strategies, including participating in affiliating business groups (Kim and Song 2017) and outsourcing logistics (Yang et al. 2016). In this article, we explore the advent of online third-party (3P) marketplaces to overcome such challenges in the Chinese market.

China's modern retail segment is primarily ecommerce based, with the online share of national chain retail expected to expand from 36.6% to 63.9% in the five years leading up to 2023 (Statista 2019b). China's giant 3P platforms such as Alibaba and JD.com (Lavin 2019) have become the engine behind

the fast growth of ecommerce in China. For example, Alibaba's Tmall, the largest online 3P marketplace in China, already boasts a 62% share of China's ecommerce market. With this 3P format, manufacturers sell directly to consumers through the marketplace website for a fee, and they make decisions independently regarding key factors such as retail prices without having to make investments in retail space or ecommerce websites (Abhishek, Jerath, and Zhang 2016). At first glance, online marketplaces seem to offer an effective solution for overcoming institutional voids and infrastructure weaknesses because marketplaces function as the market-facilitating formal institution that encapsulates all the hassles, resulting in a "hands-free" platform for manufacturers (Hagiu and Wright 2015a; Jung, Ugboma, and Liow 2015). Nevertheless, 3P online marketplaces may also bring about new challenges for manufacturers. Given most manufacturers' relatively short history of (directly) interacting with consumers and

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(independently) setting up retail space and tactics (e.g., retail prices), online marketplaces may leave manufacturers in a vulnerable position while increasing the risk of channel conflict and confusion (Tsay and Agrawal 2004). Especially in a fast-growing market such as China, not being able to rely on trusted resellers may generate substantial risks.

So far, manufacturers have been reluctant to disclose information on 3P marketplace performance in China, thereby preventing researchers from studying the potential of this platform. As a result, it remains unclear whether and to what extent joining online 3P marketplaces in China can generate market value. Anecdotal evidence suggests a mixed success rate, as examples of both unexplained success and failure abound. Procter & Gamble, for example, opened eight more stores for its brands on Tmall after its debut in 2009. In contrast, Coach entered Tmall to sell luxury handbags a few years ago and quietly closed the store in 2016 (Abkowitz and Lin 2016). Therefore, this study addresses the market value that results from adopting online 3P marketplaces in China.

Using a contingency framework, we argue that the market value of online 3P marketplaces accruing to manufacturers depends on the fit between the context (i.e., the Chinese market) and the manufacturer's market knowledge and marketing capabilities in the Chinese market. The absence of reliable sources of market information or knowledge is one of the major sources of institutional voids in emerging markets (Khanna and Palepu 2010). Without sufficient market knowledge, firms directly interacting with Chinese consumers may substantially reduce the impact of 3P platforms. Marketing capabilities, on the other hand, are the valuable, nonsubstitutable, and inimitable sources of advantage that can help a firm mitigate infrastructure weaknesses (Morgan et al. 2009) and may therefore increase the value of operating on 3P platforms.

To add an extra layer of complexity, a firm operating in a foreign market encounters additional costs that a local firm would not incur (Johanson and Vahlne 2009). Foreign firms' unfamiliarity with local Chinese consumers and their lack of roots in the Chinese environment can cause a liability of foreignness when interacting directly with such consumers through 3P platforms. Against this setting, the following questions can be raised:

- Whether and to what extent online 3P marketplaces can generate market value for consumer goods manufacturers;
- Whether and to what extent the impact is contingent on manufacturers' market knowledge and marketing capabilities; and
- Whether the market value varies for local and foreign firms.

This study contributes to the literature in several ways. At a broad level, our empirical findings, particularly regarding whether and how 3P marketplaces benefit manufacturers from various industries, complement the rich stream of theoretical literature on marketplaces (e.g., see Abhishek, Jerath, and

Zhang 2016). Specifically, our study is useful to manufacturers because it provides clear guidance for them to assess under what circumstance marketplaces may create substantive synergy. More importantly, as our study focuses on the largest online marketplace in emerging markets, Alibaba's Tmall, the insights from this research may not only be useful for brands operating in emerging markets but may also allow for reverse marketing learnings in developed markets. Indeed, as online markets are further penetrating developed markets and China's platform giants are eyeing footholds in these markets, knowing how to benefit from these emerging market players' platform operations may help brand manufacturers successfully deal with these operators outside of emerging markets. Using experience gained on the Tmall platform may thus help manufacturers establish stronger positions within online marketplaces globally.

To formulate answers to our research questions, we assess the net impact of adopting 3P marketplaces on a firm's stock market return using the announcements of store openings on Alibaba's Tmall.com in China by 408 firms in 39 industries originating from 20 countries between 2008 and 2015.

Conceptual Framework

Typically, consumer goods manufacturers can sell to consumers through three models: through retailers such as Amazon.com and Walmart ("reselling" model), through their own brand stores or sites such as the Apple Store and Nike.com ("direct model"), or through online 3P marketplace platforms such as Alibaba's Tmall ("marketplace" model). Unlike a reselling model in which retailers take ownership and control over products, both 3P marketplaces and direct models give manufacturers control. The direct and marketplace models also differ drastically in terms of operation. In direct models, manufacturers must do everything themselves, from building a retail space to acquiring new consumers and from maintaining the retail space to handling logistics. In contrast, manufacturers in 3P marketplaces share a small portion of their revenues with the marketplace and the marketplace helps them accomplish the same tasks at a lower cost by exploiting the economies of scale (Hagiu and Wright 2015b). Table 1 presents a comparison of the three models. Figure 1 depicts the conceptual framework.

Creating Market Value Through Online 3P Marketplaces

Manufacturers often find it difficult to maintain a sustained competitive advantage in emerging markets where institutional voids (Khanna and Palepu 1997) and infrastructure weakness (Vanhonacker 1997) abound. We next explore how online 3P marketplaces may assist manufacturers in these two areas.

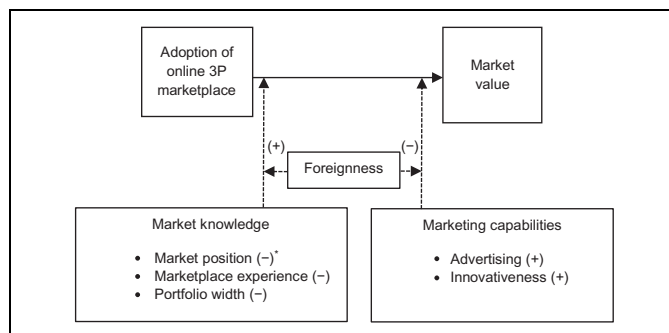
Filling institutional voids. Institutional voids reflect prevailing conditions that disturb the functioning of markets, disallow buyers and sellers to come together, and increase the transaction costs of the market activity (Doh et al. 2017). Insufficient

Table 1. Three Models of Selling to Consumers.

	Reselling	Marketplace	Direct
Control rights			
Manufacturers fully control direct access to consumers	x	✓	✓
Manufacturers fully control marketing mix	x	✓	✓
Manufacturers fully control product de/listing	x	✓	✓
Manufacturers fully control store layout	x	✓	✓
Operational responsibilities			
Manufacturers must build retail space (by themselves)	x	x	✓
Manufacturers must acquire new consumers (by themselves)	x	x	✓
Manufacturers must maintain retail space (by themselves)	x	x	✓
Manufacturers must handle logistics (by themselves)	x	x	✓

✓ indicates manufacturers have full control or take full responsibility.

x indicates manufacturers do not have full control or do not take full responsibility.

**Figure 1.** Conceptual framework. The sign between parentheses reflects the expected effect on performance.

market knowledge substantially increases the risk of operating in markets characterized by these voids. Khanna and Palepu (2010) point out that an absence or unreliable source of market information is one of the three main sources of market failure, making both foreign and domestic firms reluctant to do business in emerging markets. Online 3P marketplaces may, however, help manufacturers increase their market knowledge. In 3P marketplaces, manufacturers sell to and interact with consumers directly (Hagiu and Wright 2015b). This allows manufacturers to observe consumers and attain valuable customer information by themselves, whereas, in a reselling format, the retailers secure all this information. Of course, customer information does not necessarily equal consumer knowledge. However, marketplaces like Alibaba's Tmall provide manufacturers with services to correctly "decode" consumer behavior (Swanson 2015). Moreover, manufacturers can hone their knowledge

about products in online 3P marketplaces, as marketplace consumers can rate and leave feedback on each product. The product information collected through word-of-mouth networks can also deepen manufacturers' understanding for all products and facilitate product innovation in the future (Kozinets et al. 2010).

In contrast, online 3P marketplaces may prevent manufacturers from filling the institutional voids. Retaining both channels (online marketplaces and selling) establishes the manufacturer as a direct competitor to its reseller partner, potentially leading to tension referred to as "channel conflict" (Tsay and Agrawal 2004). Channel conflict can undermine attempts to develop cooperative relationships in the intermediated channel. Retailers, for example, may refuse to share category information with manufacturers that opened marketplace stores (Ganesan et al. 2009). This, in turn, amplifies the transaction costs that manufacturers incur in emerging markets. Although online marketplaces aid manufacturers in obtaining new market knowledge, the newly acquired knowledge may not be adequate to compensate for the potential information loss due to channel conflicts.

Overcoming infrastructure weaknesses. Infrastructure weaknesses reflect the scarcity of specialized intermediaries and distribution networks in emerging markets (Khanna and Palepu 2006). Logistics networks in emerging markets are different from those of developed countries in many ways due to regulatory, cultural, and economic factors. Unlike in developed countries, the majority of people in emerging markets live in rural areas. In China, for example, 41% of its population lives in rural areas, whereas that number is only 18% for the United States (World Bank 2018). As a consequence of the large population in rural areas, manufacturers may be faced with continuously escalating inventory holding and transportation costs to meet the expectations of these customers (Neuwirth 2014). Moreover, emerging markets often have poorly maintained transportation infrastructure. The transportation system in China is less developed than that of the United States. Navigating the underdeveloped logistics in China is not easy for manufacturers, as they have to deal with issues (e.g., difficulties in estimating delivery time and costs) that do not exist in developed markets (Jung, Ugboma, and Liow 2015).

3P marketplaces can alleviate this paucity of distribution and logistics in emerging markets. Alibaba, for example, established its own logistics collaboration platform (Cainiao), which connects three million couriers and warehouses covering 30 million square meters (Ouyang 2018). By taking advantage of the marketplace's distribution networks, manufacturers can reduce capital investment in logistics facilities, reduce inventory costs, and focus on core activities such as product development.

Still, manufacturers may fail to mitigate the negative consequences of infrastructure weakness by simply relying on online 3P marketplaces. When manufacturers outsource logistics services to online marketplaces, they unavoidably increase their dependency on marketplaces platforms. In turn,

marketplaces gain power over manufacturers and therefore are able to demand concessions from manufacturers (Gielens et al. 2018). Although online marketplaces aid manufacturers by reducing supply chain costs, the decrease in these costs may be not sufficient to cover the increase in concessions due to the power shift. Given that 3P market adoption can have both negative and positive conditions, we propose two alternative hypotheses:

H_{1a}: Online marketplace adoption generates positive market value for manufacturers.

H_{1b}: Online marketplace adoption generates negative market value for manufacturers.

Contingency Framework

In the spirit of contingency theory (Drazin and Van de Ven 1985; Luthans and Stewart 1977), we propose that the resulting market value from 3P platform adoption will be determined by the fit between the firm and the specific context at hand. Specifically, we consider two sets of contingency factors: market knowledge and marketing capabilities. Lack of market knowledge is one of the major sources of institutional voids in emerging markets (Khanna and Palepu 2010). To what extent a manufacturer can reap higher market value by using online 3P marketplaces in China depends on the fit between the manufacturer's current level of market knowledge and the knowledge contingency. That is, when 3P marketplaces can fill the institutional voids caused by insufficient market information, a manufacturer with less market knowledge may generate more value through 3P marketplace adoption.

Marketing capabilities such as product management, pricing, selling, marketing communication, and channel management represent rare, valuable, nonsubstitutable, and inimitable sources of advantage that can lead to superior firm performance (Morgan et al. 2009). To what extent a manufacturer can reap higher market value by using online 3P marketplaces in China depends on the fit between the manufacturer's current level of marketing capabilities and the capability contingency. That is, when marketplaces are able to mitigate the infrastructure weaknesses in emerging markets, a manufacturer with weaker marketing capabilities may generate more market value through 3P platform adoption.

Market Knowledge: Filling Voids

Market knowledge is knowledge that is specific to a local market regarding its language, culture, politics, society, and economy (Lord and Ranft 2000). We distinguish three sources of market knowledge in this setting: (1) market position, (2) marketplace experience, and (3) product portfolio.

Market position. Market position refers to the market share of a firm in the local market (Shepherd 1972). A superior market position is an important indicator of a firm's knowledge about the local market. The logic behind this is that a firm with broad market knowledge has heterogeneous information and

understanding about customers and competitors, enabling it to provide products and services that match the diverse needs of its customer segments, therefore achieving a superior market position (Kogut and Zander 1992). As such, a firm with a strong market position is less likely to rely on external institutions to gain market knowledge. When online 3P marketplaces are able to provide manufacturers access to market knowledge, a firm with a strong market position may not derive much added value. We thus propose that:

H_{2a}: Market value derived from adopting online 3P marketplaces is lower for firms with a strong market position.

Marketplace experience. One of the great benefits of 3P marketplace channels is that manufacturers interact with consumers, which is impossible in a reselling channel (Corsten and Kumar 2005). Once a manufacturer debuts on a 3P marketplace, it starts to accumulate knowledge about local consumers and markets as well as experience with respect to market operations, distribution systems, and regulatory practices. Over time, these experiences accumulate into tacit knowledge and expertise that can be translated to operations in other nondirect and nondigital channels (Malerba 1992). When online 3P marketplaces are able to provide manufacturers access to market knowledge, the marginal value of joining more than one 3P marketplace may be lower for manufacturers that have already reaped the benefits of working through such marketplaces. Therefore, when establishing a store on a new 3P marketplace platform, a firm with prior marketplace experience will find it less beneficial. Thus,

H_{2b}: Market value derived from adopting online marketplaces is lower for firms with prior marketplace experience.

Product portfolio. More diversified firms are more likely to maintain a rich mix of different knowledge stocks, which enriches their ability to recombine knowledge in related fields in a more complex and creative manner (Prabhu, Chandy, and Ellis 2005). A firm that has a wider product portfolio may thus gain and maintain a competitive advantage because of this knowledge spillover (Rothaermel et al. 2006). Therefore, firms with rich product portfolios can identify and target finer segments more effectively and satisfy customers' varying needs and wants more precisely. In addition to knowledge spillovers, a large portfolio of products also helps firms achieve economies of scale and scope in marketing and distribution, thus reducing unit costs (Varadarajan 1986). In summary, more diversified firms may gain less from working with 3P platforms. Therefore:

H_{2c}: Market value derived from adopting online marketplaces is lower for firms with diversified portfolios.

Marketing Capabilities: Fit with Infrastructure

Marketing capabilities refer to capabilities that are adaptive and enable the firm to adjust its strategies to fit fast-changing

markets (Day 2011). From the eight distinct marketing capabilities (Vorhies and Morgan 2005), we consider two—advertising and innovativeness—that are highly relevant to firm performance in emerging markets (Swami and Dutta 2010).

Advertising. Advertising plays an important role in increasing brand awareness (Yoo, Donthu, and Lee 2000) and creating a pull effect in the market. The insufficient infrastructure in emerging markets, however, makes the availability of advertised products extremely low in the rural areas of such markets, thereby countering and even nullifying the pull effect. Marketplaces allow consumers in emerging markets, regardless of their geographic locations, more easy access to the products they learn about through ads. Heavy advertisers may thus benefit more from joining marketplaces. We thus expect that:

H_{2a}: Market value derived from adopting online marketplaces is higher for heavy advertisers.

Innovativeness. Understanding consumer heterogeneity is of critical importance for developing new products. Successful innovation is only possible if manufacturers can obtain trustworthy and relevant information regarding the market and consumers (Adner and Levinthal 2001). However, the inadequate infrastructure in emerging markets makes it hard for manufacturers to gain and translate information from more rural areas, regardless of whether they rely on retailers or gather it themselves. 3P marketplaces solve this problem by providing manufacturers with not only a robust and reliable distribution infrastructure but also other auxiliary support, including data warehousing and data analytics services. Therefore, highly innovative manufacturers can obtain more value from working with 3P platforms. Therefore:

H_{2c}: Market value derived from adopting online marketplaces is higher for innovative firms.

The Liability of Foreignness

Lack of market knowledge is much less of an issue for local firms, as such firms are generally familiar with local markets and consumers (Makino and Delios 1996). Foreign firms, in contrast, rarely find it painless to understand local consumers.

Given that market knowledge is more of an issue for foreign firms than local firms, we expect the three contingency factors regarding market knowledge (i.e., market position, marketplace experience, and product portfolio) to affect local firms and foreign firms to different extents. The liability of foreignness causes an extra layer of knowledge scarcity, preventing foreign firms from decoding and translating information and making the adoption of marketplaces more beneficial to foreign suppliers than local ones, even when such firms have already accumulated previous knowledge (Makino and Delios 1996). Therefore, we expect the negative main effects of market knowledge to be attenuated for foreign firms and the three interaction terms with foreignness to be positive. Thus:

H_{3a}: The negative effects of market knowledge on market value are attenuated for foreign firms.

Infrastructure weaknesses can also be more pronounced for foreign than local firms. Due to cultural differences, foreign firms often need help from local partners with local knowledge to conduct business in the local environment (Inkpen and Beamish 1997). When infrastructure weaknesses are salient, foreign firms are more likely than their local counterparts to depend on 3P marketplaces to reach and understand consumers in rural areas. In turn, it becomes fairly easy for 3P marketplaces to demand concessions from foreign firms, resulting in higher transaction costs. As such, we expect the positive main effects of marketing capabilities to be attenuated for foreign brand manufacturers.

H_{3b}: The positive effects of marketing capabilities on market value are attenuated for foreign firms.

Method

Setting

We used a quasi-natural experiment examining brand manufacturers joining Alibaba's Tmall.com in China. We believe this setting offered potential to derive interesting insights for two reasons. First, online consumer growth in China's Tier 1 and Tier 2 metropolitan cities like Beijing and Shanghai has reached a plateau. More than half of online consumers—namely, 670 million—live in China's lower-tier cities and small towns (McKinsey 2019). Serving these rural consumers has been a long-standing challenge (due to, e.g., poor logistics networks) for many firms. 3P marketplaces provide lower-tier cities accessibility to firms that lack a solid footprint in rural areas. Therefore, it is not too surprising to see that leading marketplaces Tmall and JD.com have established a combined market share of over 86% in China (Statista 2019a). Second, the richness of the Tmall platform allows for generalizable insights. Since its introduction in 2008, numerous manufacturers from various categories have been attracted by the opportunities Tmall provides and have opened stores on Tmall. So far, Tmall hosts more than two million brands from more than 30 industries (at the two-digit SIC level) and serves more than 690 million active buyers (CGTN 2019; Statista 2019c).

Measuring Market Value

Two particular challenges when measuring the performance impact of joining online marketplaces are that (1) manufacturers do not disclose information on revenues or costs resulting from the marketplace entry and (2) a temporal asymmetry exists between the investment and its potential payoff (see, e.g., Borah and Tellis 2014; Geyskens, Gielens, and Dekimpe 2002 for a similar reasoning). To overcome these issues, we relied on the well-established event study method and used abnormal returns around the announcement date to infer the financial implications of manufacturers adopting a marketplace format. By using an event study methodology, we assumed that

the impact of new public information on future firm cash flows was immediately captured by changes in the equity price of the focal manufacturer. Thus, we used abnormal stock returns as the focal performance metric (Swaminathan and Moorman 2009). This performance metric is forward-looking, integrates multiple dimensions of performance, and is less easily manipulated by managers than other measures (Geyskens, Gielens, and Dekimpe 2002). Given that manufacturers who announced joining Tmall originate from multiple countries, the typical market model in a single-country setting may no longer be appropriate (Beckers, Connor, and Curds 1996). We followed Park (2004) and made use of the world market model to obtain the estimate of the expected returns by adding a world market factor into the model. This approach allowed us to derive the standardized cumulated abnormal returns (SCAR_{it}), which represent the market value that can be uniquely attributed to the adoption of an online 3P platform (see Appendix A for more information about this approach).

Sample Composition

Following recent research on event studies (Fang, Lee, and Yang 2015; Homburg, Vollmayr, and Hahn 2014), we followed four steps to define the sample. First, we identified all public firms that launched a store on Tmall from 2008 to 2015 by manually researching each store in 20 categories¹ on Tmall.com. In addition, in our sample, we only kept public firms that, in their home country, are traded on a stock exchange that belongs to the World Federation of Exchanges.² Membership in the World Federation of Exchanges requires the stock exchange to be significant within its country of origin, to be regulated by a supervisory body, to be within a statutory framework, and to be a public good (World Exchanges 2017), which ensures that the member stock exchanges demonstrate leadership in the public and are regulated marketplaces. The stock exchanges we maintained in our sample are listed in Web Appendix W1. In addition, we excluded all cross-listed equities and retained the stock price in the firm's home country. For example, China United Network Communications Group is listed on three public stock exchanges: Shanghai Stock Exchange, Hong Kong Stock Exchange, and New York Stock Exchange. Because China is the home country, we retained China United's listing on the Shanghai Stock Exchange and excluded the listings on both the Hong Kong Stock Exchange and New York Stock Exchange.

Since accurate event dates are critical to ensure a valid event study (Brown and Warner 1985), we retrieved the event date of the announcements through two sources: (1) disclosures, press releases, and articles obtained from the Factiva and LexisNexis databases, which provide access to extensive documents from various news and business sources as well as company website searches and (2) the Internet Archive Digital Library ([https://](https://archive.org/web)

archive.org/web), which provides access to historical versions of websites and allowed us to track when a website was created (Babić Rosario et al. 2016). After comparing the event date from the two sources, we retained whichever was earlier as the final event date. As a result, we compiled a sample of 501 announcements.

Next, we cross-validated the events for which a possible confounding event occurred within 20 days of the announcement of the Tmall store opening by searching on Factiva and LexisNexis. Confounding events included such activities as stock splits and structural stock changes, damage suits, product recalls, dividends, joint venture announcements, and merger and acquisition activities. Moreover, if a firm launched more than one store on Tmall, we only retained the earliest observation to avoid confounding effects of earlier events on later events. We retained 440 announcements using these criteria. To ensure that equity data were available for the 250 trading days required in the market model, we cross-checked the firm in the DataStream database and found data to be available for 416 announcements.

Finally, research in marketing (Aksoy et al. 2008) and finance (Cowan 1992) suggests that abnormal returns can be sensitive to the presence of outliers. Consistent with this line of research, we trimmed the top and bottom 1% of the abnormal returns to ensure that extreme outliers did not influence our results. After applying this criterion, the final sample consisted of 408 announcements from firms in 39 industries (at the two-digit SIC level) between 2008 and 2015 across 20 countries or regions.

The manufacturers in our sample are active in various categories, such as automobiles (e.g., Ford, Toyota, Volkswagen), consumer electronics (e.g., Apple, Lenovo, Philips), personal care (e.g., L'Oréal, Henkel, Procter & Gamble), apparel (e.g., GAP, Hugo Boss, VF Corporation), packaged food (e.g., China National Cereals, Oils and Foodstuffs Corporation, General Mills, Kellogg's), alcoholic drinks (e.g., Budweiser, Diageo, Kweichow Moutai), and pharmaceuticals (e.g., Harbin Pharmaceutical Group, Pfizer, Johnson & Johnson). The sample spans well-known, billion-dollar companies like Unilever and Royal Dutch Shell but also covers a variety of smaller, unknown suppliers such as Joyoung and Septwolves. The countries of origin in our sample represent almost 75% of the world's economy and are graphically depicted in Figure 2.

For all these manufacturers, we further collected information on different knowledge and capability constructs. Specifically, we measured market position as a firm's weighted market share of all the brands it sells in China. 3P marketplace experience is a dummy variable that equals 1 when the firm previously entered JD.com, the second largest marketplace in China, before entering Tmall, and 0 otherwise. We measured a firm's portfolio width as the number of categories in which the manufacturer operates in China. Advertising is a firm's total annual advertising expenditure in China. We measured innovativeness as the logarithm of the firm's annual (global) research and development expenditures (Banerjee et al. 2015). All independent variables were measured in the year prior to Tmall

¹ See Tmall's category definition at http://about.tmall.com/tmall/fee_schedule.

² <http://www.world-exchanges.org/home/>

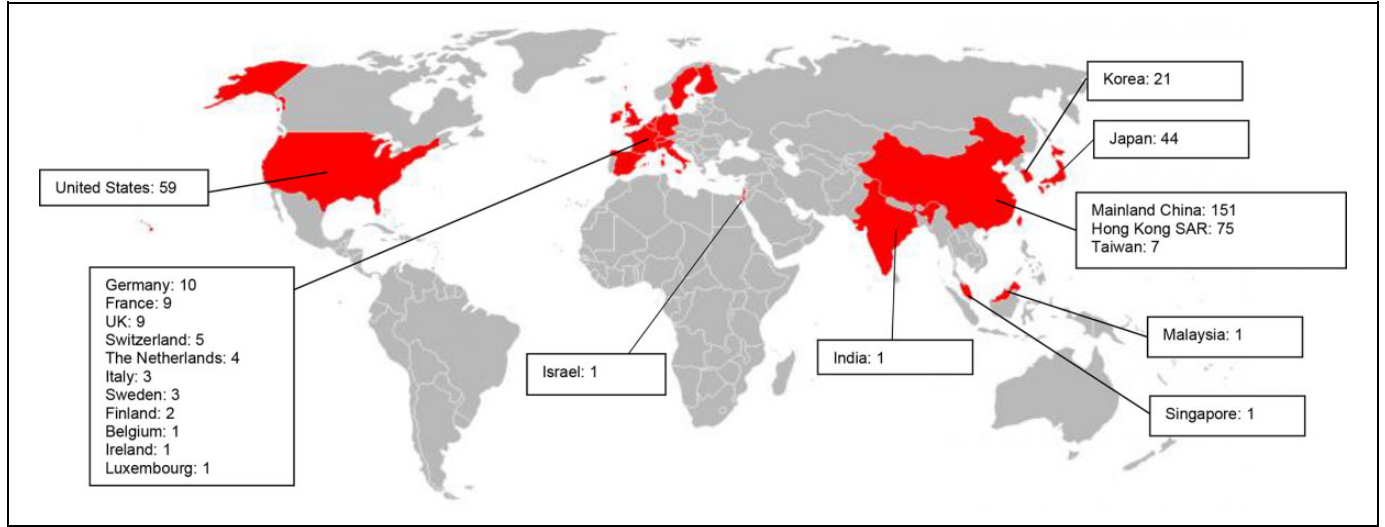


Figure 2. Countries represented in the sample and number of firms in each country.

adoption. Table 2 provides a more extensive overview of the operationalization of the various measures, and Table 3 summarizes the descriptive statistics and correlations of the variables.

Hypotheses Testing

To test our hypotheses, we regressed the SCARs on our set of substantive predictors and control variables:

$$\begin{aligned}
 \text{SCAR}_{i,t} = & \gamma_0 + \gamma_1 \text{MKTPOS}_{i,t} + \gamma_2 \text{CHLEXP}_{i,t} \\
 & + \gamma_3 \text{PORWID}_{i,t} + \gamma_4 (\text{FOREIGN}_{i,t} \\
 & \times \text{MKTPOS}_{i,t}) + \gamma_5 (\text{FOREIGN}_{i,t} \\
 & \times \text{CHLEXP}_{i,t}) + \gamma_6 (\text{FOREIGN}_{i,t} \\
 & \times \text{PORWID}_{i,t}) + \gamma_7 \text{ADV}_{i,t} + \gamma_8 \text{INV}_{i,t} \\
 & + \gamma_9 (\text{FOREIGN}_{i,t} \times \text{ADV}_{i,t}) \\
 & + \gamma_{10} (\text{FOREIGN}_{i,t} \times \text{INV}_{i,t}) \\
 & + \gamma_{11} \text{FOREIGN}_{i,t} + \gamma_{12} \text{AOT}_{i,t} \\
 & + \gamma_{13} \text{ROA}_{i,t} + \gamma_{14} \text{LVG}_{i,t} + \gamma_{15} \text{SEV}_{i,t} \\
 & + \gamma_{16} \text{GDP}_{i,t} + \gamma_{17} \text{IMR}_{i,t} \\
 & + \sum_{l=1}^L \gamma_{17+l} \text{Year}_l + \varepsilon_{i,t}, \quad (1)
 \end{aligned}$$

where MKTPOS, CHLEXP, and PORWID reflect the experience constructs market position in China, marketplace experience, and product portfolio, respectively. ADV represents the firm's advertising in the Chinese market, and INV represents innovativeness. FOREIGN is a dummy variable that equals 1 if the firm does not originate from China. We further controlled for other variables that might affect abnormal stock returns. More specifically, we included total assets (TOA), return on assets (ROA), and financial leverage (LVG) (Dhaliwal 1986) because the adoption of the marketplace format may be more valuable for more profitable firms (Homburg,

Vollmayr, and Hahn 2014). At the category level, we captured whether the firm operates in a service industry rather than in consumer product industries by including a service dummy variable (SEV). Finally, we augmented the model with country- and year-specific influences by including the gross domestic product (GDP) in the year of opening the store, store opening year dummies (YEAR), and the inverse Mills ratio (IMR). All continuous independent variables were mean centered to ease interpretation except for the inverse Mills ratio. To correct for a potential violation of the statistical independence assumption that exists because some firms have the same country of origin, we clustered standard errors at the country level (Auerbach and Hassett 2005). The error term $\varepsilon_{i,t}$ follows a normal distribution. In estimating this model, we may encounter a potential selection bias. Firms may have relevant private information that is not fully known to the market but that influences their decisions to join a marketplace in an emerging market. However, we can only observe firms that actually announced store establishment at Tmall in public media sources (i.e., $\text{Open_Store}_{i,t} = 1$). To account for this potential selection effect, we use a two-stage Heckman selection model (Heckman 1977; see Appendix B for more information on this procedure).

Results

Does 3P Marketplace Adoption Influence Market Value?

Analysis of the daily abnormal returns for 10 days around the event, as described in Table 4, reveals that the day after the announcement exhibits a significant positive stock market reaction ($p < .05$). The effect on the day of the announcement itself is not significant. This is, however, encountered frequently in international settings where time differences must be factored in (see, e.g., Park 2004). The abnormal return on the day before the announcement is also not significant,

Table 2. Summary of Measures and Data Sources.

Variable (label)	Data Source	Measure ^a
Market position (MKTPOS)	Euromonitor, Mintel Oxygen	A firm's weighted market share in China (range is [0, 1]), whereby the weight is the percentage of a firm's total sales in the category.
Marketplace experience (CHLEXP)	Factiva, Firm websites, Internet Archive Digital Library	Dummy variable that equals 1 when a firm launches a marketplace store on JD.COM, and 0 otherwise.
Portfolio width (PORWID)	Euromonitor, Mintel Oxygen	Number of categories that a firm operates in China.
Advertising (ADV)	Kantar Media	Annual advertising expenditure in China (unit: Million Chinese Yuan).
Innovativeness (INV)	Compustat	Logarithm of annual R&D expenditure (unit: Million U.S. Dollar); Compustat item: XRD.
Foreign firm (FOREIGN)	Compustat	Dummy variable that equals 1 when a firm does not originate from China, and 0 otherwise; Compustat item: LOC.
Total assets (TOA)	Compustat	Annual total assets; Compustat item: AT.
Return on assets (ROA)	Compustat	A firm's net income divided by total assets; Compustat items: NI/AT.
Financial leverage (LVG)	Compustat	A firm's total liabilities divided by shareholder equity; Compustat items: LT/SEQ.
Service (SEV)	Compustat	Dummy variable that equals 1 when a firm's two-digit SIC code is between 70 and 89, and 0 otherwise.
GDP of home country (GDP)	Compustat	GDP of a firm's home country <i>j</i> .
SGA (SGA)	Compustat	A firm's sales and general administrative expenses (SGA); Compustat item: SGA.
Adoption rate (ADPRAT) ^b	Compustat	Percentage of firms from the focal firm's home country <i>j</i> that adopted Tmall.

^aAll variables are measured in the year prior to the event year.

^bVariable used exclusively in the selection equation.

Table 3. Descriptive Statistics (N = 408).

	Mean ^a	SD	1	2	3	4	5	6	7	8	9	10	11
Market position (MKTPOS)	.04	.07	1.000										
Marketplace experience (CHLEXP)	.45	—	-.017	1.000									
Portfolio width (PORWID)	2.22	2.92	.425	-.051	1.000								
Advertising (ADV)	2.29	7.64	.325	-.027	-.042	1.000							
Innovativeness (INV)	54.02	754.38	.103	.010	.239	-.007	1.000						
Foreign firm (FOREIGN)	.45	—	.009	-.209	.187	-.019	.461	1.000					
Total assets (TOA)	59.91	74.06	.159	-.213	.058	.271	.059	.236	1.000				
Return on assets (ROA)	8.92	9.50	.062	.151	.089	.074	-.053	-.009	-.071	1.000			
Financial leverage (LVG)	.029	.11	-.029	-.064	.097	-.045	.239	.220	-.020	.008	1.000		
Service (SEV)	.09	—	.177	-.233	-.129	.085	-.145	-.130	.367	-.087	-.071	1.000	
GDP of home country (GDP)	4.39	5467.58	-.019	-.165	.009	-.021	.025	.463	.133	.090	-.109	-.030	1.000

^aFor dummy variables, we report the proportion of observations that have a value of one.

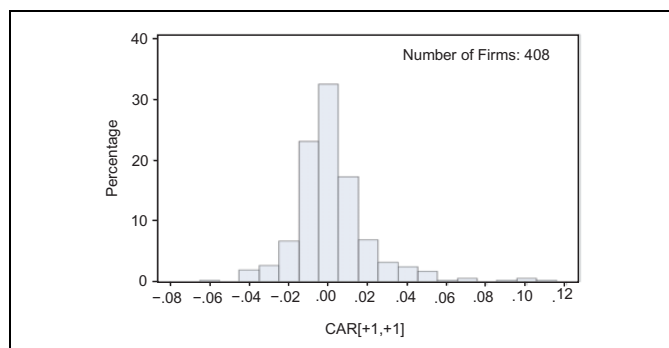
indicating that there is no evidence of information leakage before the announcement. With a positive abnormal stock return of .13% ($p < .05$) on average, performance is enhanced when a manufacturer joins a 3P marketplace platform in China, lending support for H_{1a} . This implies that with an average increase of .13% in the cumulative abnormal return, the shareholder value of a median-sized manufacturer (based on manufacturers' market capitalization on the day before the announcement) with a market capitalization of \$2.141 billion is augmented (adjusted for overall market movements) by \$3 million in one day. Stock markets thus acknowledge that marketplaces can effectively help manufacturers to address the potential challenges in China.

Although the abnormal returns are, on average, significant and positive for manufacturers, stock markets recognize that not all manufacturers can reap benefits from adopting 3P marketplaces to the same extent, as is reflected by the substantial variation in cumulative abnormal returns (CARs) as shown in Figure 3. Whereas 49% reported a positive effect, 51% experienced a negative outcome. Some manufacturers that gained the most are Nongshim and Gree Electric, with CARs of 3.0% and 2.2%, respectively. With a market capitalization of \$264 million, Nongshim, for instance, gained \$8 million in shareholder value. Still, some of the biggest losers were Mengniu Dairy and Li-Ning, with CARs amounting to -1.8% and -1.3% and who lost about \$156 million and \$52 million, respectively, in shareholder

Table 4. Abnormal Stock Returns.

Average Abnormal Returns (AAR _{<i>t</i>})				
Event Day	Mean AR	Percentage Positive AR	Patell <i>t</i>	
−10	−.09%	40.69%	−.83	
−9	.03%	48.28%	.63	
−8	−.16%	45.83%	−1.31	
−7	.13%	50.00%	1.52	
−6	−.02%	47.55%	−.48	
−5	−.02%	49.76%	−.16	
−4	.01%	51.23%	.53	
−3	−.14%	43.63%	−1.15	
−2	.17%	51.96%	1.46	
−1	−.14%	42.16%	−1.49	
0	−.09%	43.63%	−.80	
1	.13%	49.27%	2.03**	
2	.00%	45.10%	.52	
3	−.24%	42.40%	−1.20	
4	−.04%	45.83%	−.23	
5	.06%	48.28%	.85	
6	−.09%	45.59%	−1.12	
7	−.14%	43.63%	−1.73	
8	−.02%	48.04%	.34	
9	−.02%	48.28%	.4	
10	.01%	45.83%	.59	
Cumulative Average Abnormal Returns (CAAR _[−<i>t</i>1, <i>t</i>2])				
Event Day	Mean CAR	Mean SCAR	Percentage Positive CAR	Patell <i>t</i>
−10 to +10 days	−.07%	−1.78%	48.04%	−.36
−3 to +3 days	−.29%	−.98%	45.48%	−.20
−3 to +2 days	−.06%	1.04%	47.19%	.21
−3 to +1 days	−.09%	−.47%	46.70%	−.10
−2 to +2 days	.08%	3.99%	50.00%	.81
−2 to +1 days	.07%	2.65%	48.17%	.55
−1 to +1 days	−.10%	−1.03%	45.48%	−.21
0 to +1 days	.03%	3.64%	49.39%	.74
0 to +2 days	.05%	5.06%	50.12%	1.02
0 to +3 days	−.18%	1.80%	46.94%	.36

*** $p < .01$, ** $p < .05$, * $p < .10$.

**Figure 3.** Distribution of cumulative abnormal returns.

value. To understand the heterogeneity in manufacturers' CARs, we relate them to the set of knowledge and marketing strength constructs as detailed in Equation 1.

Do Contingency Factors Influence the Value Creation of 3P Marketplace Adoption?

Table 5 reports the results of our contingency analysis. We also report the relative effect sizes for all estimates (Steenkamp et al. 1999). The correlations between all variables are below the recommended threshold of .8 (Judge et al. 1988), and none of the VIFs exceed 10 (maximum VIF = 2.08) (Hair et al. 2010). Thus, multicollinearity is not likely to be a problem. The estimate of the inverse Mills ratio is significant ($\gamma_7 = .130$, $p < .05$), implying that a selection correction is indeed warranted. The positive sign suggests that, on average, unobserved factors that make a firm more likely to open a store on Tmall tend to have a positive effect on shareholder value implications.

With respect to market knowledge and experience, the results support H_{2a} and H_{2c}. It indicates a negative and

Table 5. Regression Results (N = 408).

Dependent Variables: SCAR[+I,+I]	Expectation	Estimate	t-value	Relative Effect Size
Intercept		.392	1.45	
Market knowledge				
Market position (MKTPOS)	—	-.699	-4.37***	.0002
Marketplace experience (CHLEXP)	—	.304	4.04***	.0091
Portfolio width (PORWID)	—	-.031	-4.20***	.0021
Market position × foreign firm	+	1.875	2.12**	.0039
Marketplace experience × foreign firm	+	-.06	-.45	.0002
Portfolio width × foreign firm	+	.023	1.2	.0007
Marketing Capabilities				
Advertising (ADV)	+	.019	20.30***	.0129
Innovativeness (INV)	+	-.031	-1.79*	.0029
Advertising × foreign firm	—	-.036	-3.34***	.0183
Innovativeness × foreign firm	—	.026	1.78*	.0018
Control Variables				
Foreign firm (FOREIGN)		-.165	-1.23	.0021
Total assets (TOA)		-.008	-.44	.0002
Return on assets (ROA)		-.01	-2.81**	.001
Financial leverage (LVG)		.293	2.14**	.0009
Service (SEV)		.004	.31	.0001
Home country's GDP (GDP)		-.009	-.38	.0007
Year fixed effects (YEAR)		Included		
Selection Correction				
Inverse mills ratio (IMR)		.13	1.97**	
R ²		.077		

*** $p < .01$, ** $p < .05$, * $p < .10$

significant main effect ($\gamma_1 = -.699$, $p < .01$) for market position and for product portfolio width ($\gamma_3 = -.031$, $p < .01$), suggesting that less diversified firms can benefit more from marketplaces. However, a positive effect ($\gamma_2 = .304$, $p < .01$) is reported for 3P marketplace experience on firm returns. Therefore, H_{2b} is not supported. Firms that have prior experience with a 3P marketplace channel can reap more benefits when joining a new marketplace. A potential explanation is that, when joining a new 3P marketplace, a firm with prior marketplace experience has a learning-by-doing advantage (Argote and Ingram 2000). Previous learnings from one 3P marketplace can provide manufacturers with a roadmap of pitfalls to avoid, which reduces the inherent risk of entering new marketplaces and helps manufacturers make appropriate choices for sourcing, production, marketing, organizational, and other activities when entering a subsequent marketplace.

Regarding foreignness, the results lend support for its moderating effect on market position ($\gamma_4 = 1.875$, $p < .05$) but not on 3P marketplace experience ($\gamma_5 = -.060$, $p > .10$) or portfolio width ($\gamma_6 = .023$, $p > .10$). Thus, for foreign firms with a strong market position in the host country, the value created by adopting 3P marketplaces is stronger, thereby indicating that a certain level of knowledge is required for foreign firms to be capable of decoding information and to reap benefits from marketplaces.

With respect to marketing capabilities, we find that firms that advertise substantively in the Chinese market can benefit more ($\gamma_7 = .019$, $p < .01$), which supports H_{2d} . In addition, in contrast to our expectations, we find a negative effect of R&D

($\gamma_8 = -1.79$, $p < .10$). Thus, H_{2e} is not supported. This result may be explained as follows. When a manufacturer heavily relies on new products, more retailer support may be needed. New products represent a risk to consumers (Arndt 1967) that retailers may help to overcome. The retailer's decision to list and thus procure the new product may signal a certain level of trust in the success of the new product. By operating on 3P marketplaces, the manufacturer forgoes this external seal of retailer commitment and thus potentially lowers the success of new products.

Regarding the moderating effects of foreignness, the results indicate that foreign firms that invest less in advertising and therefore have lower brand awareness in the host country can benefit more from joining a marketplace, as indicated by the negative and significant effect ($\gamma_9 = -.036$, $p < .01$). In contrast to our expectations, we find that an innovative, foreign firm may benefit more from the consumer knowledge marketplaces offer when the firm adopts them ($\gamma_{10} = .026$, $p < .10$).

The relative effect sizes indicate higher than average effects³ for all main effects and two interaction terms: advertising and innovativeness. Specifically, for local firms, advertising, marketplace experience, and portfolio width matter the most, whereas for foreign firms, advertising, market position, and innovativeness are more important.

To gain a better understanding of the moderating impact of foreignness, we plot the significant interactions in Figure 4

³ With 25 parameter estimates, the average relative effect size is .04.

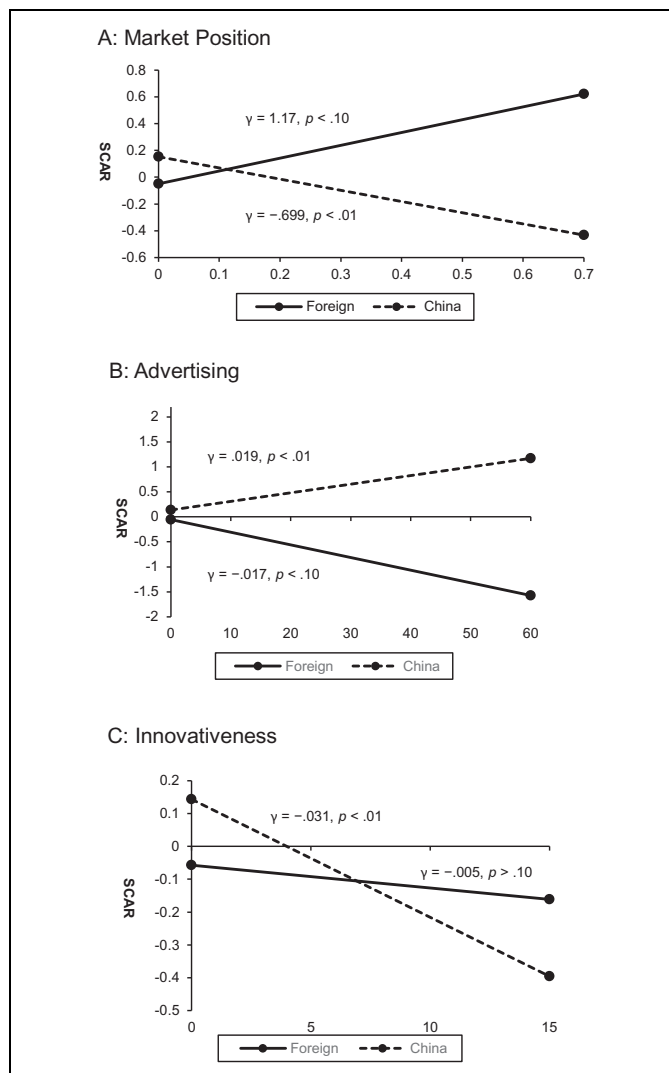


Figure 4. The moderating effects of liability of foreignness.

(Cohen et al. 2003). As can be seen in Figure 4, Panel A, the inverse association between a local firm's experience depth and market value ($\gamma = -.699, p < .01$) is reversed for foreign firms ($\gamma = 1.17, p < .10$). Local firms that hold a market position of more than .2 even stand to incur market value losses. This finding indicates that, *ceteris paribus*, a foreign firm that already has a deeper understanding of the local market can achieve higher predicted market value and thus financially benefit more than its local rivals when adopting a 3P marketplace.

As for advertising, our results indicate that the effect diverges for foreign firms and local firms (see Figure 4, Panel B). Local firms that are heavily involved in advertising can reap more benefits when joining marketplaces ($\gamma = .019, p < .01$). In contrast, this effect reverses for foreign firms: Foreign firms who spend less on advertising benefit more from adopting a 3P marketplace format ($\gamma = -.017, p < .10$).

Again, the impact of R&D diverges for foreign and local firms. Figure 4, Panel C, indicates that local firms that are more

innovative benefit less from adopting marketplaces ($\gamma = -.031, p < .10$) and may even stand to lose if their R&D spending exceeds more than \$54.4 million. In contrast, the negative effect of innovativeness on market value is nullified for foreign firms ($\gamma = -.005, p > .10$).

Robustness Checks

To further increase confidence in the results, we conducted various robustness checks. First, we conducted tests to verify whether the positive evaluation of marketplace adoption was a temporary reaction that was quickly corrected afterward. Specifically, we used a graphical test by plotting CARs over days before or after the announcements and a pooled regression of the abnormal returns on the number of trading days (up to 100 days) since the announcement. Second, we verified the robustness of our results regarding the calculation of abnormal returns and alternative event windows.

Table 6 summarizes all the concerns that may have potentially affected our findings, along with details on how we assessed their robustness. In addition, Web Appendix Tables W2–W4 and Figure W1 exhibit the results of each robustness check. Small numerical differences notwithstanding, all our reported results are substantively robust.

Discussion

Due to the largely unorganized retail structure in China's markets, global manufacturers often find it impossible to simply lean on retailers to reach all potential consumers, rural or urban, the way they are accustomed to doing in developed economies. The advent of online 3P marketplaces, however, provides manufacturers with a new model to tackle the infrastructure and knowledge challenges in the Chinese market. Unsurprisingly, many manufacturers are flocking to this model, which offers them the promise of tapping directly into the vast Chinese market without having to make upfront investments in logistics or infrastructure.

On paper, this model seems to be a winning proposition, especially given its increasing popularity among consumers in China. However, online 3P marketplaces still involve direct distribution, which is a function and skill very few consumer goods manufacturers have mastered so far, especially when targeting large-scale access in virtually untapped markets. To learn whether and to what extent consumer goods manufacturers stand to gain from working through marketplaces in China, we collected data on over 400 Chinese and foreign manufacturers opening a storefront on China's largest online marketplace, Tmall, and assessed to what extent these firms' market value changed following the announcement. To further gain insight into which firms may experience gains, we used a contingency framework and related manufacturers' short-term abnormal returns to manufacturers' market knowledge and marketing capabilities. The findings provide comprehensive guidance for manufacturers, both global and Chinese, to assess whether and to what extent they can take advantage of online 3P marketplaces to thrive in China.

Table 6. Robustness Checks.

Concern	Proposed Robustness Check	Findings
Whether the positive evaluation of marketplace adoption is not just a temporary reaction that is quickly corrected afterward.	We plotted CAARs over the days before or after the announcements. We estimated a pooled regression of the abnormal returns on the number of trading days (up to 100 days) since the announcement.	CAARs stayed at a higher level after the event, indicating that the positive evaluation is not just a short-term lift that evaporates in the days following the announcement. See Figure W1. This estimation shows no significant drift (i.e., $b_{\text{time}} = .00$, $p = .87$), indicating that the initial positive evaluation is not merely a short-term reaction that is corrected in the subsequent weeks (Geyskens, Gielens, and Dekimpe 2002; Homburg, Vollmayr, and Hahn 2014).
Whether the results remain stable for alternative calculations of the abnormal stock returns.	We reestimated the models with abnormal returns calculated by the market model, which a method that has been widely used in prior studies (Geyskens, Gielens, and Dekimpe 2002). We reestimated the models with unstandardized abnormal stock returns as the dependent variable (Swaminathan and Moorman 2009).	The results remain stable in terms of directions and significance. See Appendix W2. The results remain stable in terms of directions and significance. See Appendix W3.
Whether the results remain stable for alternative event windows.	We reestimated the models with the following alternative windows: (1) Day 1 to Day 2, (2) Day 1 to Day 3, and (3) Day 0 to Day 2.	The results remain stable in terms of directions and significance. See Appendix W4.

Theoretical Contributions

This study contributes to marketing theory regarding online marketplaces in two main ways. First, the direct effect of marketing knowledge on firm performance is not monotonically positive as was expected in prior literature (Luca and Atuahene-Gima 2007). Instead, in the context of 3P marketplaces, we see a stunning difference between traditional market knowledge (sourcing from market position and portfolio diversification) and digital knowledge (sourcing from marketplaces experiences). Our results show that online marketplaces can compensate for an inadequacy of traditional market knowledge but not digital knowledge. The latter can only be acquired through learning-by-doing (Argote and Ingram 2000). This indicates the importance of classifying the source of market knowledge in marketing theory.

Moreover, our results indicate that highly advertised (local) firms can gain more benefits from marketplaces, but highly innovative (local) firms would gain less. This is in line with prior research suggesting that firms tend to gain more when they have an emphasis on value appropriation (i.e., advertising) but less when they have an emphasis on value creation (i.e., innovation) (Mizik and Jacobson 2003). However, we find that these insights flip for foreign firms. This shows that the country of origin may be another important (but ignored) moderator of the relationship between strategic marketing emphasis and firm performance.

Last but not least, the extant theory shows that foreignness is a liability. Our results demonstrate that this is not necessarily the case in online marketplaces. When online 3P marketplaces function as an intermediary to facilitate knowledge acquisition, foreign firms can gain more benefits from working with the

intermediary because they have many more learning opportunities than their local counterparts. Once again, this finding emphasizes the multifaceted roles of foreignness.

Managerial Implications

Do firms benefit from joining marketplaces in China? Although we find an average positive CAR, indicating that firms can gain from online marketplaces, the substantial variation in CARs clearly indicates that online marketplaces are not a one-size-fits-all solution for all companies handling the daunting challenges in China. We find that the performance implications of adopting marketplaces are largely contingent on manufacturers' market knowledge and strengths.

Is knowledge golden in the Chinese market?

When scrutinizing the issue through the lens of the knowledge-based view (Grant 1996), we expected that more knowledgeable firms would require less external infrastructure as offered by 3P marketplaces to thrive. We find that both Chinese firms with a less established market position and firms with less extensive product portfolios, irrespective of origin, reap more benefits when working with marketplaces, indicating that marketplaces may help these manufacturers overcome some of the knowledge hurdles in the Chinese market. Common sense may suggest that manufacturers with wider portfolios are more experienced and fare better than specialized manufacturers in dynamic environments. On the contrary, our findings show that marketplaces allow specialized manufacturers to closely observe consumers and understand the market dynamism to compensate for their inadequacy of broad knowledge. In addition, manufacturers with more marketplace

experience can substantively benefit more from joining a new marketplace. The learnings gleaned from joining one marketplace channel can significantly increase the efficiencies and reduce the uncertainties associated with joining new marketplace operations. Indeed, the extent to which manufacturers can achieve greater gains in financial performance depends on how fast they can learn to create synergy with online marketplaces, and the previous learnings from one marketplace increase the tendency for learning curves to “plateau” or level off (Epple, Argote, and Devadas 1991). We, therefore, recommend that firms not rush into a giant marketplace like Tmall but to first accumulate some operational experience in other 3P marketplace channels.

Can marketplaces prevent marketing capabilities from turning into weaknesses in the Chinese market?

We find that heavy advertisers may benefit from joining a marketplace, but only when the firm is a Chinese player. Foreign firms that spend less on advertising benefit more from adopting a marketplace format than foreign firms who spend heavily. Thus, a foreign firm with a lower advertising budget will generate less brand awareness but can use a marketplace as a powerful tool to compensate for this weakness. In contrast, a Chinese firm that heavily invests in advertising is more likely to cultivate higher brand equity and can utilize a marketplace to preserve its brand equity.

Can marketplaces galvanize R&D in the Chinese market?

Marketplaces provide manufacturers with the access to consumers needed to collect consumer information, which is a necessity for product innovation, and thus may enable more innovative firms to galvanize their R&D inputs. However, we find that this does not hold for foreign and Chinese firms alike. Foreign firms’ performance outcomes from joining a marketplace do not appear to be influenced by their innovativeness. Chinese firms that rely heavily on new products may benefit more from retailers rather than marketplaces, as retailers may help to build trust. In addition, Chinese firms may have stronger relationships with local retailers, thereby allowing them to leverage retailer support more to their advantage. Forsaking this support in favor of marketplaces may thus harm local, innovative firms.

Is foreignness a liability on Chinese 3P marketplaces?

The answer is not unqualifiedly positive. Unlike previous international business studies (e.g., Johanson and Vahlne 2009; Ojala and Tyrväinen 2007) that conclude that foreign firms are in an inferior position when competing with local firms, we find that foreign firms do not necessarily in all cases gain less from working with online marketplaces compared with their local counterparts. One possible reason for this is that online marketplaces are a brand new model for every firm, regardless of origin. Since foreign and Chinese firms actually start on the same level, both face potentially steep learning curves (Yelle 1979). Therefore, the extent to which manufacturers can achieve greater financial gains depends on how fast they can learn to create synergy with online marketplaces through channel and product experience, or brand equity, regardless of origin. Still, when relying heavily on advertising to communicate

with Chinese consumers, foreignness can indeed be a liability when opening a store on a marketplace.

Limitations and Further Research

This study and the interpretation of the results must be considered while taking into account various general and special limitations, some of which suggest possible avenues for future study. First, we use shareholder value as the focal performance metric. Although shareholder value is generally used in marketing–finance interface studies (Swaminathan and Moorman 2009), it is not without limitations (Homburg, Vollmayr, and Hahn 2014). Therefore, future work may consider other metrics, such as sales and profit, to evaluate the impact of joining a marketplace.

In addition, although this study accounts for market knowledge and capability contingencies, further important moderators may be of central interest in managerial decisions. For example, a firm’s operational experience (global vs. regional) and brand positioning (global vs. local) may affect the performance of expansion in different markets (Meyer 2001; Townsend et al. 2009). Furthermore, previous studies (Roth 1995) have found culture and socioeconomics to influence consumers’ perceptions of country of origin and firms in foreign markets. Therefore, an investigation of the impact of underlying cultural, economic, and administrative differences between the home country and host country may be another fruitful area for future studies.

Finally, further research could also examine the potential for channel conflict due to the adoption of the marketplace. How will the existing channel members react to the new channels? What kind of remedy can the manufacturers take to mitigate the channel conflicts? This line of exploration may further contribute to the successful development of marketplace operations in the long run.

Appendix A

To capture market value through an event study approach, we followed Park (2004) and used the world market model to obtain the estimate of the expected returns by adding a world market factor into the model⁴:

$$R_{ijt} = \alpha_{ij} + \beta_{ij} R_{mjt} + \gamma_i R_{wmt} + \mu_{ijt}, \quad (A1)$$

where R_{ijt} is firm i ’s stock return in its home country j on day t , R_{mjt} is the market index return in country j on day t , and R_{wmt} is the world market index return on day t . The parameter estimates α , β , and γ were obtained by an ordinary least squares regressing R_{ijt} on R_{mjt} and R_{wmt} by using the daily stock return data for each firm over a period of 250 trading days, ranging from 260 to 10 calendar days prior to the event day (Swaminathan and Moorman 2009). We used the MSCI World Market

⁴ Note that we could not use a Fama-French approach because Fama-French factors are only available for the American stock market.

Index as R_{wmt} , which covers 1,612 large and midcap stocks across 23 countries (Harvey 1991). We then used the estimates obtained from this model to predict the daily abnormal returns for firm i originating from country j at time t as

$$AR_{ijt} = R_{ijt} - E(R_{ijt}) = R_{ijt} - (\alpha_{ij} + \beta_{ij} R_{mjt} + \gamma_i R_{wmt}). \quad (A2)$$

The effect of joining a marketplace may be spread over a couple of days, especially in a multiple-country setting. First, possible leakage of information prior to the event day and gradual dissipation of information post event occur at various speeds in different countries. Second, the asynchronism in stock market trading hours between the home and host countries prevents potential investors from simultaneously responding to major announcements. Third, the stock and local markets in different countries may have different schedules for such events because of national holidays or regional celebrations. To account for this possible gradual dissipation of information, we calculated the daily abnormal returns of a firm i from home country j over a time window $[t_1, t_2]$:

$$CAR_{ij}[-t_1, t_2] = \sum_{t=-t_1}^{t_2} AR_{ij}. \quad (A3)$$

The selection of the appropriate event window is empirically determined on the basis of the significance of the CARs for various event windows surrounding the event day, beginning ten days before the announcement and ending ten days after the announcement. We chose the event windows with the most significant Patell t -statistics (Swaminathan and Moorman 2009). Moreover, to correct for the firms' differences in variance in daily stock prices, we standardized the cumulated abnormal returns (SCAR_{*i*}) by using the standard deviation of abnormal returns over the estimation window (Geyskens, Gielens, and Dekimpe 2002).

Appendix B

To account for potential selection issues, we used a two-stage Heckman selection model (Heckman 1977) and estimated the following binary probit regression to explain the probability of opening a store on Tmall:

$$\begin{aligned} \Pr(\text{Open_Store}_{i,t}) = & \lambda_0 + \lambda_1 \text{CHLEXP}_{i,t} + \lambda_2 \text{SGA}_{i,t} \\ & + \lambda_3 \text{INV}_{i,t} + \lambda_4 \text{ADPRAT}_{i,t} \\ & + \lambda_5 \text{FOREIGN}_{i,t} + \lambda_6 \text{TOA}_{i,t} \\ & + \lambda_7 \text{ROA}_{i,t} + \lambda_8 \text{LVG}_{i,t} + \lambda_9 \text{SEV}_{i,t} \\ & + \sum_{l=1}^L \lambda_{9+l} \text{Year}_l + \zeta_{i,t}. \end{aligned} \quad (B1)$$

We included an exclusion variable that was expected to affect a brand manufacturer's decision to open a store on a marketplace but would not affect performance. Specifically,

we add the cumulative adoption rate (ADPRAT), which represents the percentage of firms originating from the same home country as the focal firm that have already opened a Tmall store. To meet the relevance criterion, we argue that firms may mimic the adoption decisions of peers in their own country (Gielens and Dekimpe 2007). To meet the exclusion restriction, we argue that peer firms collectively either cannot observe or measure the focal firm's omitted variable(s) or cannot act on those variable(s) strategically (Germann et al. 2015).

In addition, we included the same variables in the selection model as in Equation 1, unless the required information was not available for the suppliers that uniquely feature in the selection sample (see, e.g., Robinson, Tuli, and Kohli 2015 or Swaminathan and Moorman 2009 for a similar practice). In addition, we used a manufacturer's selling, general, and administrative expenses (SGA) as a proxy measure for its advertising and its brand equity, as this construct was available for all firms in the selection sample (Gielens et al. 2018). Furthermore, we added year-fixed effects to control for time fluctuations. To ensure temporal separation between the independent variables and the adoption decision, we used one year lagged independent variables.

We estimated Equation B1 using a sample of 1,571 firms. We included firms that were a reasonable proxy for our set of focal firms by selecting proxy firms whose size ranged within $\pm 50\%$ of the focal firm's size and that operated in the same industry (based on the four-digit SIC code) (Swaminathan and Moorman 2009). All proxy firms were evaluated in the same year as the focal firm's decision to open a store on Tmall. We drew on the parameters to compute the inverse Mills ratio (IMR), which we added as a regressor to Equation 1.

Table B1 presents the results for the selection model. The likelihood ratio test shows good model fit ($\chi^2(15) = 1107.05$, $p < .001$). The hit rate of 94.9% is significantly better than chance ($61.5\% = \alpha^2 + (1 - \alpha)^2$, with $\alpha = 26.0\%$; Morrison

Table B1. Selection Model Results (N = 1,570).

Selection Model Component	Coefficient	Standard Errors
Intercept	-2.299	.202***
Instrument		
Adoption rate (ADPRAT)	2.877	.235***
Regressors of Equation 1		
Marketplace experience (CHLEXP)	6.861	166.8
Advertising (in thousands) (SGA)	.0011	.00045**
Innovativeness (in thousands) (INV)	.0083	.0005*
Foreign firm (FOREIGN)	-.332	.118***
Total assets (in millions) (TOA)	.0007	.0002***
Return on assets (ROA)	.0015	.013
Financial leverage (LVG)	.301	.213
Service (SEV)	-.538	.208***
Home country's GDP (GDP)	.063	.023***
Year fixed effects (YEAR)	Included	
Hit rate		95.00%
LR- χ^2		1114.85***

*** $p < .01$, ** $p < .05$, * $p < .10$.

1969). The exclusion restriction variable is significant and positive, thereby indicating that firms mimic the adoption decisions of peers in their own country ($p < .01$).

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